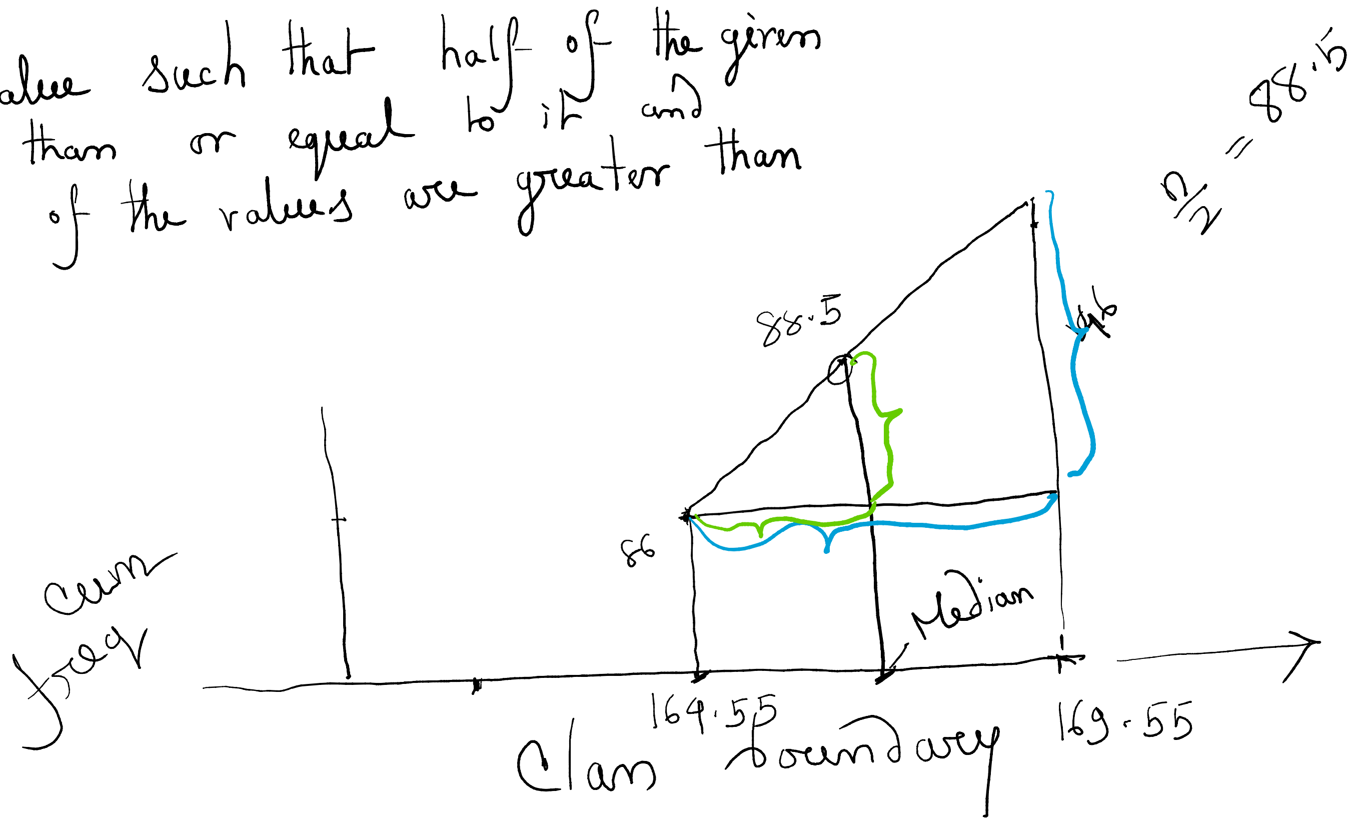
PRST (MCC 505) 3/09/2020

Class interval	frequency	Cumulative frequency
144.55 − 149.55	1	1
149.55 − 154.55	3	4
154.55 − 159.55	24	28
159.55 − 164.55	58	86
164.55 − 169.55	60	146
169.55 − 174.55	27	173
174.55 − 179.55	2	175
179.55 − 184.55	2	177
total	177	

$\frac{n}{2} = 88.5$

Median is the value such that half of the given values are less than or equal to it and remaining half of the values are greater than or equal to it.



$$\left(\frac{169.55 - 164.55}{146 - 86} = \frac{\text{Median} - 164.55}{88.5 - 86}\right)$$

Quantile (of order p)

$Z_p \quad 0 < p < 1$

→ is the value such that a proportion p of the total number of given observations are less than or equal to it and proportion (1−p) ~~of the~~ are greater than or equal to it.

$x_1, \ldots \quad \ldots \quad x_n$

$Q = (n+1).P$

if Q is integer $\quad Z_p = x_Q$

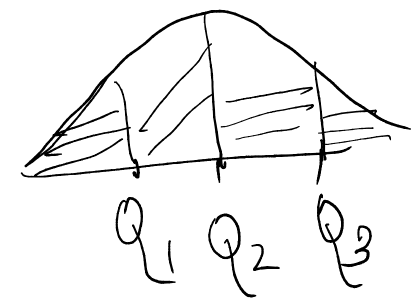
Q is not integer, $\quad r = [Q]$ $\quad$ | $Q = 2.3$, $r = 2$

$Z_p = x_r + (Q-r).(x_{r+1} - x_r)$

$p = \frac{1}{4} \quad Z_{\frac{1}{4}} = Q_1$ first quartile

$Z_{\frac{1}{2}} = Q_2$ 2nd "

$Z_{\frac{3}{4}} = Q_3$ 3rd "



$p = \frac{i}{100} \quad i = 1, 2, \ldots, 99$

$\hookrightarrow$ ith percentile

Measure of dispersion :

Data set 1	Data set 2
10, 0, 5, 5, 30	8, 9, 10, 11, 12
mean $= 10$	mean $= 10$
range $= 30 - 0 = 30$	range $= 12 - 8 = 4$

Data set 1	Data set 2
10, 11, 11.5, 11.6, 20	10, 12, 15, 18, 20
range $= 10$	range $= 10$

Variance $\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2$, standard deviation $= \sqrt{\text{var}}$

Mean absolute deviation about a pt $A = \frac{1}{n}\sum_{i=1}^{n}|x_i - A|$

n $\quad$ ≤2 D. $\quad$ A =

$$\frac{1}{N} \sum_{i=1}^{n} (x_i - \bar{x})^2 f_i \qquad N = \sum_{i=1}^{n} f_i$$

$$\frac{1}{n} \sum_{i=1}^{n} (x_i - \bar{x})^2 = \frac{1}{n} \sum_{i=1}^{n} x_i^2 - (\bar{x})^2$$

$$\frac{1}{n} \sum_{i=1}^{n} (x_i - A)^2 \geqslant \frac{1}{n} \sum (x_i - \bar{x})^2$$

$$(x_i - \bar{x} + \bar{x} - A)$$

$$\frac{1}{n} \sum |x_i - A| \geqslant \frac{1}{n} \sum |x_i - \text{Median}|$$

Moments

$m_r' = \frac{1}{n} \sum_{i=1}^{n} (x_i - A)^r$ — rth moment about the pt A.

if $A = 0$, $m_r' = \frac{1}{n} \sum x_i^r \rightarrow$ rth raw moment

$A = \bar{x}$, $m_r = \frac{1}{n} \sum (x_i - \bar{x})^r$ — rth central moment.

$m_0 = 1 \qquad m_1 = 0,$

$m_0' = 1,$

$$\boxed{\begin{aligned} &\frac{1}{n} \sum (x_i - \bar{x}) \\ &= \frac{1}{n} \sum x_i - \frac{n\bar{x}}{n} \\ &= \bar{x} - \bar{x} = 0 \end{aligned}}$$

$$m_2 = m_2' - m_1'^2$$

$$m_3 = m_3' - 3 m_2' m_1' + 2 m_1'^3$$

$$m_4 = m_4' - 4 m_3' m_1' + 6 m_2' m_1'^2 - 3 m_1'^4$$

$$(\bar{x} - A) = d = \frac{1}{n} \sum_{i=1}^{n} (x_i - A) = m_1'$$

$$m_1' = m_1 + d$$

$$m_2' = m_2 + 2 m_1 d + d^2$$

$$m_3' = m_3 + 3 m_2 d + 3 m_1 d^2 + d^3$$

$$m_4' = m_4 + 4 m_3 d + 6 m_2 d^2 + 4 m_1 d^3 + d^4$$